Eric Ding

971-348-7909 | ericding@cs.cornell.edu | ericdinging.github.io

Education

Cornell University Ithaca, NY, United States
Ph.D., Computer Science, Direction: Systems and Networking, GPA: 4.0 *Aug. 2024 – June 2029 Expected*

University of Michigan Ann Arbor, MI, United States
B.S.E., Computer Science, Summa Cum Laude, GPA: 3.96, Dual Degree Program *Aug. 2022 – Apr. 2024*

Shanghai Jiao Tong University Shanghai, China
B.S.E., Electrical and Computer Engineering, Outstanding Graduate, GPA: 3.76, top 10% *Sep. 2020 – Aug. 2024*

Research Experience

Graduate Student Researcher Ithaca, New York
Cornell University *Aug. 2024 – Present*

- Designing networking protocols and ML workload simulator for circuit-switched AI datacenter scale-out.
- Building profiling and benchmarking pipelines for evaluating collective communication implementation.
- Implemented straggler-mitigation collective communication algorithm for distributed ML workloads.
- Developed physical layer routing algorithms for silicon photonic fabrics in rack-scale ML networks.

Machine Learning System Research Assistant Ann Arbor, MI, United States
SymbioticLab, University of Michigan *May 2023 – Apr. 2024*

- Developed *Propius*, a Federated Learning (FL) resource management system, based on microservice architecture.
- Implemented scheduling policies in *Propius* that improve the average FL job completion time by up to 1.88x compared to random allocation.
- Contributed to open-source project *FedScale* in adaptive FL optimizer implementation.

Embedded System Research Assistant Ann Arbor, MI, United States
The Fan Lab, University of Michigan *May 2023 – Sep. 2023*

- Designed a reliable communication protocol that operates atop I2C and Bluetooth Low Energy protocols, ensuring high-fidelity data communication between two in-device microcontrollers and a terminal microcontroller.

Work Experience

Artificial Intelligence Internship Shanghai, China
Bosch *May 2024 – Aug. 2024*

- Designed and implemented a multi-agent system using LangGraph, OpenAI endpoints, and Neo4J graph database for retrieval augmented generation and domain-specific task reasoning.
- Deployed the system on Microsoft Azure for internal use, achieving an estimated 30% reduction in man-hour.

Publication

- **Eric Ding**, Chuhan Ouyang, Rachee Singh, Photonic Rails in ML Datacenters, Twenty-Fourth ACM Workshop on Hot Topics in Networks (HotNets 2025) (arXiv:2507.08119)
- Arjun Devraj, **Eric Ding**, Abhishek Vijaya Kumar, Robert Kleinberg, Rachee Singh, Efficient AllReduce with Stragglers, Under submission (arXiv:2505.23523)
- Abhishek Vijaya Kumar, **Eric Ding**, Arjun Devraj, Darius Bunandar, Rachee Singh, Morphlux: Transforming Torus Fabrics for Efficient Multi-tenant ML, Under submission (arXiv:2508.03674)
- **Eric Ding**, Rachee Singh, PipSwitch: A Circuit Switch Using Programmable Integrated Photonics, Optical Fiber Communication (OFC) Conference 2025 (arXiv:2501.18136)
- Jiachen Liu, Fan Lai, **Ding Ding**, Yiwen Zhang, Mosharaf Chowdhury, Venn: Resource Management Across Federated Learning Jobs, MLSys 2025 (arXiv:2312.08298)
- Anjali Devi Sivakumar, Ruchi Sharma, Chandrakalavathi Thota, **Ding Ding**, Xudong Fan,WASP: Wearable Analytical Skin Probe for Dynamic Monitoring of Transepidermal Water Loss, ACS Sensors 2023.

Volunteering

- EuroSys 2026 Shadow PC, *Oct. 2025*
- SIGCOMM 2025 AE Committee, *Aug. 2025*
- MLSys 2025 AE Reviewer, *Mar. 2025*
- Lunch and Learn Czar, *Computer Science Department, Cornell University, Feb. 2025 - May. 2025*
- Computer Science Peer Tutor, *School of Information, University of Michigan, Feb. 2024 - May. 2024*

Teaching Experience

- TA, Systems for Large-scale ML (CS5470), Cornell University, Fall 2025. Responsible for assignment development: AI inference performance benchmarking, request scheduling and batching mechanisms, CUDA kernel optimization.

Projects

A Fully Synthesizable Out-of-Order RISC-V Processor | *SystemVerilog, Verdi*

- Designed an out-of-order processor with features including: RISC-V R10k style register renaming scheme, N-way superscalar, load/store queue, tournament branch predictor, and non-blocking cache.
- Wrote the processor and a complete set of testcases in SystemVerilog, used Verdi for debugging and verification.

CNN Convolution Optimization | *CUDA, Slurm*

- Optimized forward convolution layer computation in CUDA, adopting parallel programming techniques, such as memory coalescing, shared memory multiplication and loop unrolling.
- Achieved a processing time of 0.079 seconds for large batches (10K $33 \times 33 \times 12$ images with 24 filters).

Simplified Operating System Kernel | *C++*

- Developed a CPU scheduler and a thread library, supporting thread allocation, interruption, and synchronization primitives, such as mutex and conditional variables.
- Developed a pager which manages processes' virtual address spaces, and swap-backed and file-backed pages across physical memory and disk. Built a multi-threaded network file server in UNIX file system hierarchy.

Distributed Search Engine | *Python, React, AWS*

- Developed a fault-tolerant distributed system running MapReduce framework. Created segmented inverted indexes of web pages through a MapReduce pipeline that is compatible with Hadoop Streaming.
- Implemented a distributed backend index service, capable of generating customized search results via PageRank and TF-IDF integration, and a scalable frontend search server.

Grants and Awards

- Graduate School Fellowship, Cornell University, Feb. 2024
- James B. Angell Scholar, University of Michigan, Feb. 2024
- University Honors, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Dean's List, University of Michigan, Dec. 2023, Apr. 2023, Dec. 2022
- Tang Junyuan JI Scholarship Nominee, Aug. 2022
- Shanghai Jiao Tong University Merit Student Award, Nov. 2021

Technical Skills

Languages: C++, Python, C, SystemVerilog, Go, RISC-V, Matlab, Javascript, Bash, SQL, R
Frameworks: CUDA, NCCL, PyTorch, Tensorflow, JAX, vLLM, DeepSpeed, TorchTitan, LangGhain, React, gRPC, Hadoop
Developer Tools: Git, Docker, Kubernetes, Redis, Nvidia Nsight, LaTeX, Neo4J, VSCode, Arduino, STM32CubeIDE, Wireshark, Gurobi
Simulation and Modeling: Verdi, Catia, Matlab, Mathematica, LabVIEW, Pspice, Proteus, Vivado, Astra-Sim, ns-3, gem5
Selected courses: EECS470 Computer Architecture, EECS 471 Applied Parallel Programming with GPUs, EECS482 Operating System, EECS489 Computer Network